

AI/ML intern DAY 18

Body Landmark Detection & Measurement Analysis Report

(Member-3)

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1. Introduction

Human body landmark detection is an important application of Artificial Intelligence and Computer Vision. Modern pose estimation models can identify key body joints from images and videos, allowing systems to understand body posture and estimate physical measurements.

Virtual Try-On systems rely heavily on accurate landmark detection because garment alignment and body measurement estimation depend on the precise localization of body joints.

This research focuses on studying MediaPipe Pose, evaluating landmark detection accuracy, estimating body measurements, and understanding how pose estimation improves Virtual Try-On systems.

2. Objectives

The objectives of this research are:

- Study body landmark detection.
 - Test MediaPipe Pose on different body images.
 - Analyse landmark detection accuracy.
 - Estimate body measurements.
 - Evaluate Virtual Try-On applications.
 - Document observations and findings.
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3. Understanding Body Landmark Detection

Body landmark detection is the process of identifying important body joints using Artificial Intelligence.

The landmarks detected include:

- Nose

- Eyes
- Ears
- Left Shoulder
- Right Shoulder
- Left Elbow
- Right Elbow
- Left Wrist
- Right Wrist
- Left Hip
- Right Hip
- Left Knee
- Right Knee
- Left Ankle
- Right Ankle

MediaPipe Pose detects a total of 33 body landmarks.

These landmarks provide a skeletal representation of the human body.

4. MediaPipe Pose

Overview

MediaPipe Pose is a lightweight pose estimation framework developed by Google.

It performs real-time body landmark detection using deep learning models.

Features

- Real-time processing
- 33 body landmarks
- High processing speed
- Cross-platform support
- Lightweight architecture

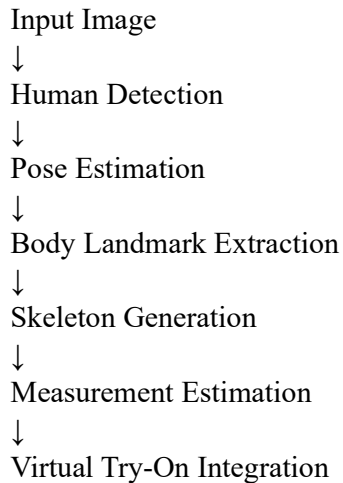
Applications

- Virtual Try-On
- Fitness tracking

- Gesture recognition
 - Motion analysis
 - Body measurement extraction
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5. Body Landmark Detection Workflow

The body landmark detection process consists of the following stages:



This workflow enables AI systems to understand human body structure.

6. Body Measurements Studied

The following body measurements were analysed.

Shoulder Width

Estimated using the left and right shoulder landmarks.

Height Estimation

Calculated using the distance between head and ankle landmarks.

Waist Position Detection

Estimated using torso and hip landmarks.

Hip Width

Calculated using left and right hip landmarks.

Arm Length

Estimated using shoulder, elbow, and wrist landmarks.

7. Experimental Analysis

MediaPipe Pose was tested using images with different body poses.

Front View

Observation:

- Excellent landmark visibility.
- Accurate shoulder detection.
- Stable body skeleton generation.

Side View

Observation:

- Slight reduction in landmark confidence.
- Hidden body parts reduce accuracy.

Arms Raised

Observation:

- Arm tracking remains stable.
- Wrist landmarks detected correctly.

Different Body Poses

Observation:

- Most landmarks remain correctly localized.
- Extreme poses introduce minor deviations.

8. Pose Accuracy Analysis

Several factors affect landmark detection accuracy.

Lighting Conditions

Good lighting improves confidence scores.

Camera Angle

Front-facing images provide better accuracy.

Image Resolution

Higher-resolution images improve landmark localization.

Body Visibility

Full-body images generate more reliable measurements.

Clothing Type

Loose garments may reduce body contour visibility.

9. Importance of Body Measurements in Virtual Try-On

Body measurements improve the quality of Virtual Try-On systems by:

- Improving garment alignment.
- Estimating user body size.
- Personalizing outfit recommendations.
- Enhancing cloth fitting.
- Reducing incorrect size selection.

Accurate measurements provide a better digital shopping experience.

10. Code Implementation

```
import cv2
import mediapipe as mp

# Load image
image = cv2.imread("person.jpg")

if image is None:
    print("person.jpg not found")
    exit()

# Initialize MediaPipe Pose
mp_pose = mp.solutions.pose
mp_draw = mp.solutions.drawing_utils

pose = mp_pose.Pose(
    static_image_mode=True,
    min_detection_confidence=0.7
```

```
)

# Convert image to RGB
rgb = cv2.cvtColor(image, cv2.COLOR_BGR2RGB)

# Process image
results = pose.process(rgb)

if results.pose_landmarks:

    # Draw landmarks
    mp_draw.draw_landmarks(
        image,
        results.pose_landmarks,
        mp_pose.POSE_CONNECTIONS
    )

    # Get shoulder coordinates
    h, w, _ = image.shape
    lm = results.pose_landmarks.landmark

    left_shoulder = lm[11]
    right_shoulder = lm[12]

    shoulder_width = abs(
        int(left_shoulder.x * w) -
        int(right_shoulder.x * w)
    )

    print(
```

```
        "Estimated Shoulder Width:",
        shoulder_width,
        "pixels"
    )

    # Save output
    cv2.imwrite(
        "landmark_output.jpg",
        image
    )

    print("Landmark Detection Completed")
    print("Output saved as landmark_output.jpg")

    cv2.imshow(
        "MediaPipe Pose",
        image
    )

    cv2.waitKey(0)
    cv2.destroyAllWindows()

else:

    print("No human detected.")
```

Output:



11. Research Findings

The study revealed that body landmark detection provides an efficient method for estimating body dimensions from a single image.

Front-facing images produced the best results, while side-view images slightly reduced accuracy.

Shoulder width estimation was found to be the most reliable measurement, whereas waist position estimation depends heavily on body visibility and clothing type.

MediaPipe Pose offers a lightweight and practical solution for Virtual Try-On applications.

12. Challenges Identified

Several challenges were observed during testing.

Occlusion

Body parts hidden by objects reduce detection quality.

Camera Perspective

Different camera angles affect landmark positioning.

Loose Clothing

Loose garments hide body contours.

Low Lighting

Poor lighting reduces landmark confidence.

Extreme Body Poses

Complex poses may introduce small localization errors.

13. Applications

Body landmark detection is widely used in:

- Virtual Try-On Systems
 - Digital Tailoring
 - Fitness Tracking
 - Healthcare Monitoring
 - Motion Capture
 - Human Activity Recognition
 - Avatar Generation
 - Fashion Technology
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14. Learning Outcomes

During this research, the following concepts were studied:

- Pose estimation
 - Body landmark detection
 - MediaPipe Pose
 - Body measurement estimation
 - Computer vision workflows
 - Virtual Try-On systems
 - Human skeletal tracking
 - AI-based body analysis
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15. Conclusion

This research explored body landmark detection and body measurement estimation using MediaPipe Pose. The study demonstrated how body landmarks can be used to estimate shoulder width, height, waist position, hip width, and arm length from a single image.

The findings indicate that accurate landmark detection significantly improves Virtual Try-On systems by enabling better garment alignment and personalized fitting experiences. MediaPipe Pose provides a fast, lightweight, and effective solution for real-time body analysis and serves as a strong foundation for future AI-powered fashion applications.